



Advanced Analytics &
Business Consultancy

Building reliable GenAI systems at LTPlabs

How to think and build GenAI



April 2025

Eduardo Castro

The launch of new products leveraging GenAI for general use has democratized access to the power of LLMs

General use GenAI chatBots



ChatGPT



Bard



perplexity



Claude

The launch of new products leveraging GenAI for general use has democratized access to the power of LLMs

General use GenAI chatBots



Around
40%
of the U.S. population age 18-64
uses GenAI monthly

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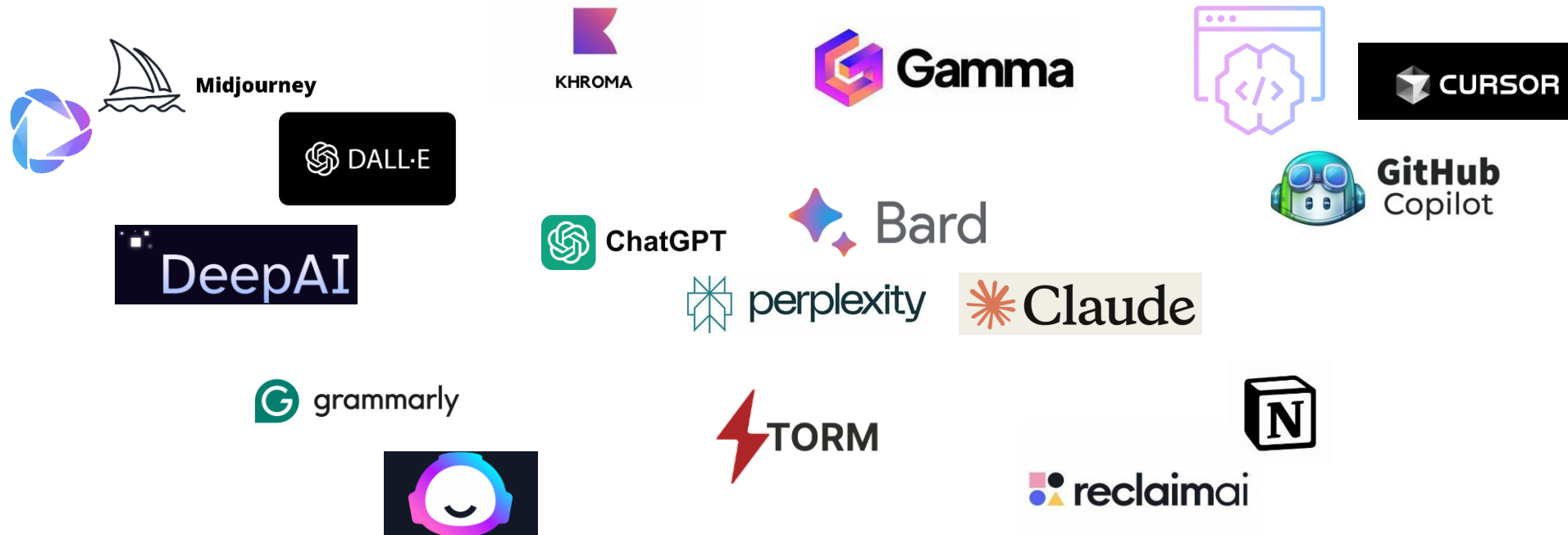


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Fast
Convenient
Do not require expertise

At the same time, we witness an explosion of new companies and tools leveraging the same technology for specific use cases

Examples of new GenAI tools



And **much much much** more...

The landscape of GenAI technology is rapidly changing with increasing adoption by companies and individuals

Timeline of the development of general use GenAI technologies

Market size of AI in 2024 \$233bn
Projected **annual growth** rate
(CAGR 2025-2032) of 29.2%

Landmark paper
"Attention is all you need"



Vaswani et al. 2017 – [Attention is all you need](#)
Statista, 24-Nov-2024 [Artificial Intelligence - Worldwide | Market Forecast](#)

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What happened here?

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Timeline of the development of general use GenAI technologies

Not just models but...

Data collection
Infrastructure
Product Development
Ethical considerations

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But what is GenAI?

One classical and one contemporaneous definition

AI – Machine Learning

Discriminative Models

- Classifiers
- Regressors

$$P(Y|X=x)$$

Generative Models

- Generative models

$$P(X)$$

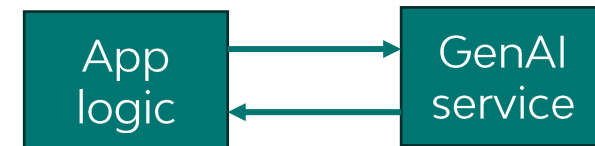
Types of Data:

- Time-series
- Images
- Text
- ...

When people talk about **GenAI** nowadays they are referring to the use of **Large Language Models**.

These models are typically trained, stored and made available by large AI companies like OpenAI, Google, Anthropic. Applications access these models as a service.

Training GPT-4 costs around \$20 million and takes around 2 months.
Unfeasible for almost all companies.



But what is GenAI?

GenAI can mean different things depending on who you are talking to

 **Academics:** learn data distributions to generate new content like text, images, or code.

 **General Public:** can write, draw, or create things based on what you ask it to do

 **Policy Makers:** generates human-like content, with implications for jobs, ethics, and regulation.

 **Business:** creates content to boost productivity and automate repetitive tasks.

 **Developers:** Models (e.g., LLMs, diffusion) that produce new outputs which are useful

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And make decisions...

Key points of difference



Business & analytics together

Our **hybrid profile** combines **advanced analytics with business expertise**, with + 200 scientific papers published

Tailored and complete solutions

We specialize in tailoring **analytical solutions that are custom-built** to meet your specific needs, **from design to execution**

Long-Term Partnership

We partner with market leaders **during every step of their analytical journey**, building value through impactful, lightning-fast initiatives

ACROSS THE WHOLE AI JOURNEY

DISCOVER

Inspire analytical journeys and craft the vision / roadmap

Screen analytical opportunities via workshops and assessments

DRIVE CHANGE

Sustain and enhance analytics products along their lifecycle

Make change happen, fostering adoption



DESIGN & DECIDE

Co-create and design **analytics solutions**

Draw on analytical approaches to **enlighten complex decisions**

DEVELOP & ACCELERATE

Build **tailored analytics products**

Accelerate analytics initiatives & empower teams

A “Pull AI” approach can increase the odds of success

Push AI (data centric)

Data-driven decision making

Anchor on data that is available

Find a purpose for data

Start from what is know

Empower data scientists

Pull AI (problem centric)

Decision-driven data analytics

Anchor on a decision to be made

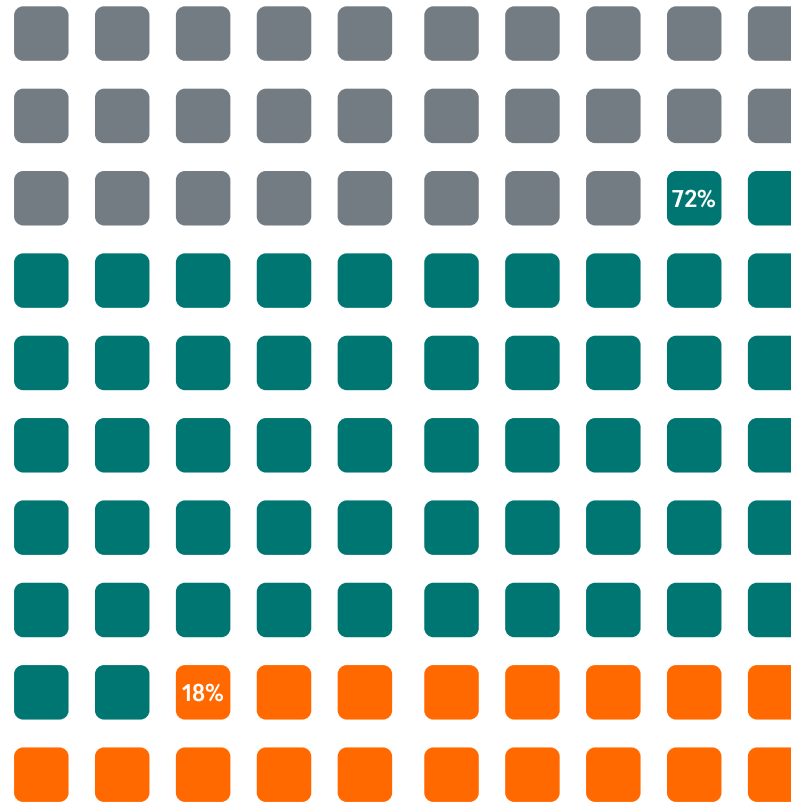
Find data for a purpose

Start from what is unknown

Empower decision makers

AI hype has outpaced its impact as this stark contrast suggests

*While 72% of companies
are investing millions in
(Gen)AI initiatives ¹*



*Only about one in four
achieve tangible value ²*

¹ <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>

² <https://www.bcg.com/press/24october2024-ai-adoption-in-2024-74-of-companies-struggle-to-achieve-and-scale-value>

GenAI can improve and automate tasks across different departments and levels within organizations

GenAI across organizations



Mostly led
by humans
**GenAI
as a tool**



Mostly led
by AI
**GenAI as
automation**

Customer service

GenAI compiles feedback, analyses sentiment and recommends improvements

GenAI copilot for support teams, suggesting solutions and paths

GenAI chatbot that handles orders, complains, and FAQs escalating to human if needed

HR and Talent management

GenAI scans CVs, ranking candidates based on HR defined criteria

GenAI reads company policy and serves as 1st level support – internal GPT

AI tracks employee efficiency, customer feedback and generate actionables

Marketing and Sales

GenAI personas helps marketing to craft and execute campaigns

GenAI generates dynamic comms to marketing team

GenAI nurtures leads in automated workflow for corp health checkups

Operations and Supply Chain

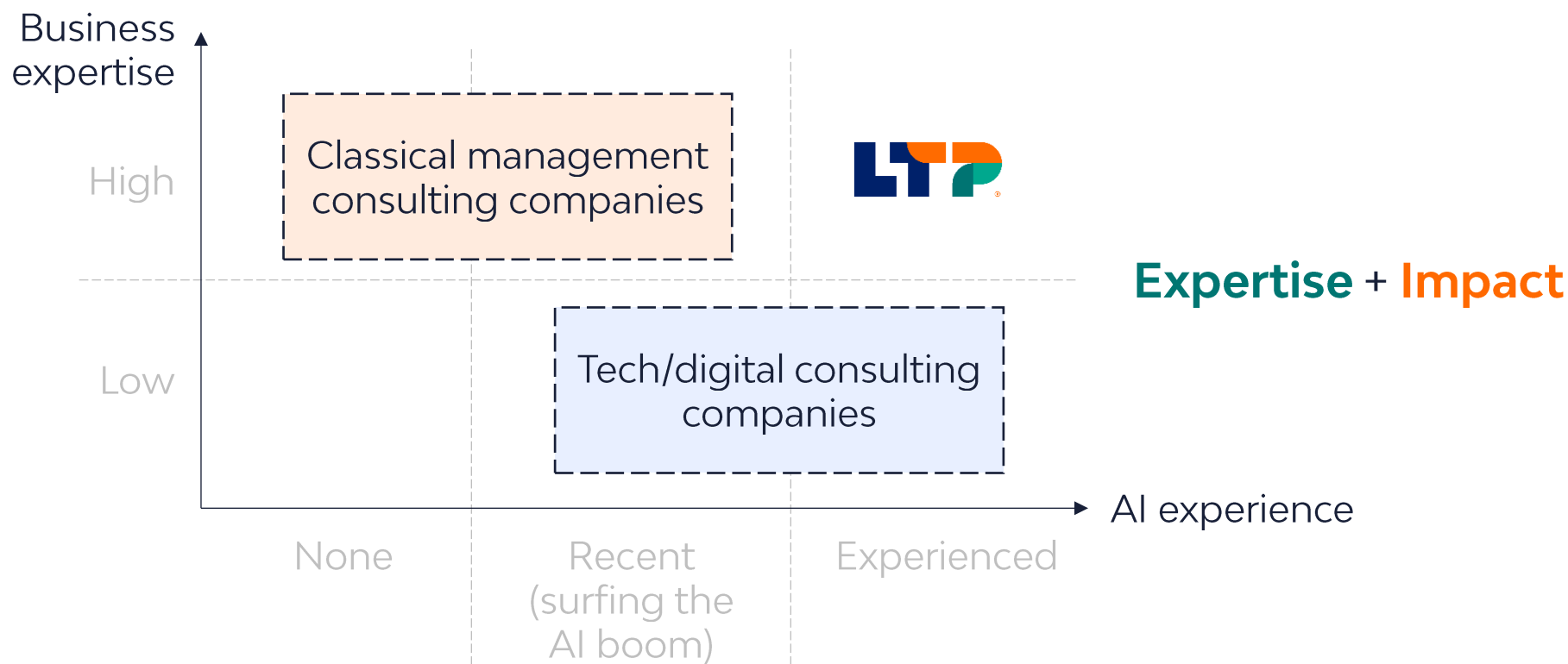
GenAI analyses clinic workflow data and suggests improvements

GenAI forecasts demand and proposes optimized slots

GenAI generated procurement plans, generating orders and negotiations

About LTPlabs - empowering business with AI & Analytics

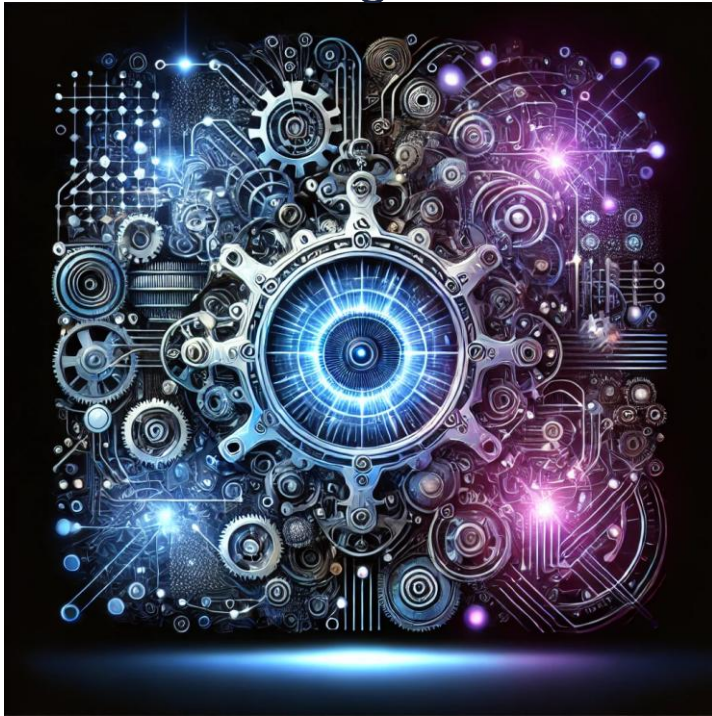
AI & Analytics | Business Expertise



Today we are focusing on three layers of the GenAI wave

Presentations structure

LLMs – the engine of GenAI



Solving complex tasks reliably



Real use-cases inside LTPlabs





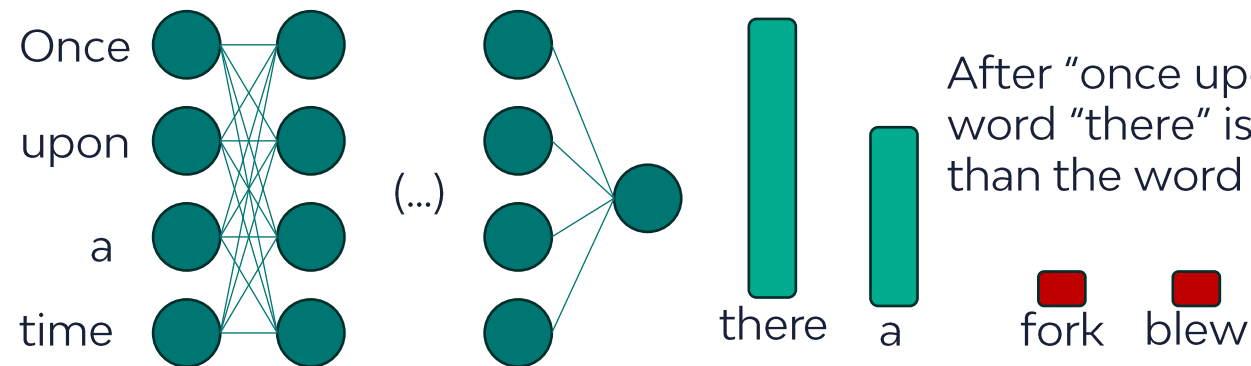
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Large Language Models

Typical approaches for natural language generation use NN for next word prediction

Neural networks for next token prediction

Text generation by sampling among the high probability words, adding them to the phrase, and running the process again.



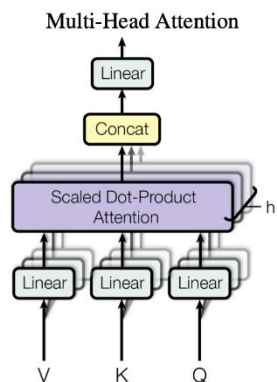
Typical approaches for natural language generation use NN for predicting the probability of the next word in sentences. Turns out **increasing model size and data** has a **huge impact** on prediction **quality**.

LTPlabs is a boutique company that provides advanced analytical methods and business flair to help our clients make better complex decisions, build more

Attention is the basic block in the architecture of LLMs, and they work by next token prediction during training and inference

Large Language Models

The attention mechanism behind LLMs



Attention allows **reasoning over relationships between tokens**, independently of their position in the sequence, contrary to CNNs and that came before. RNNs



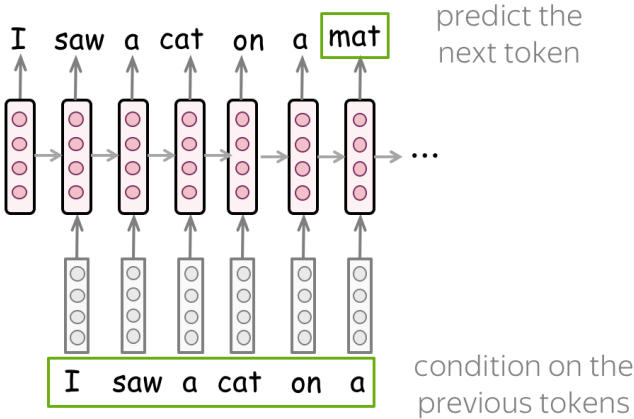
Self-supervised training through backpropagation

The first king of

the world
Portugal
England

orange
going
car

Next token prediction during inference



Multi-head attention figure adapted from Vaswani et al. 2017
Next token prediction figure adapted from Lena Voita, 2024 – [link](#)

*Typically, transformers have positional encodings enabling learning positional relationships as well

LLMs do not have a learning objective relating to “truthfulness” of their output, but rather the “plausibility” of the next token

Considerations about the use of LLMs

Hallucinations

John Locke (1632-1704) was an English philosopher and physician, widely regarded as one of the most influential of Enlightenment thinkers.

Leonardo di ser Piero da Vinci (1452-1519) was an Italian polymath of the High Renaissance who was active as a painter...

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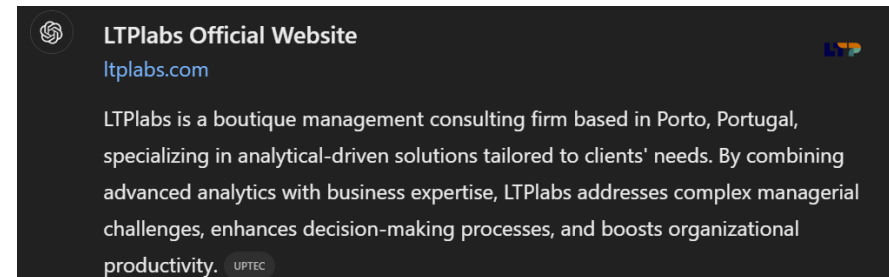
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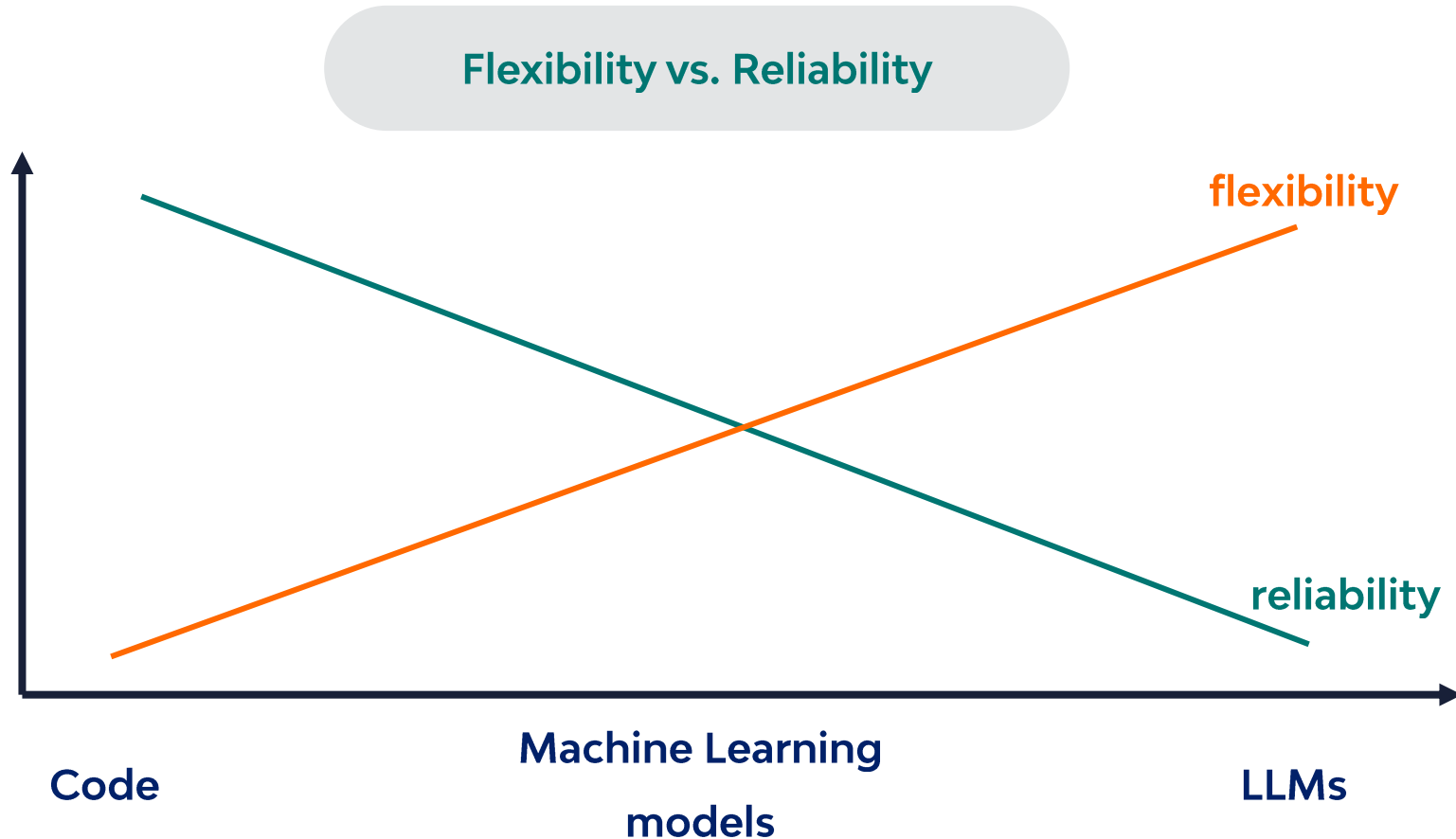
Authorship



ChatGPT outputting it after your prompt does not make you the author!!

Typically, with increased scope and flexibility, our systems become less and less reliable

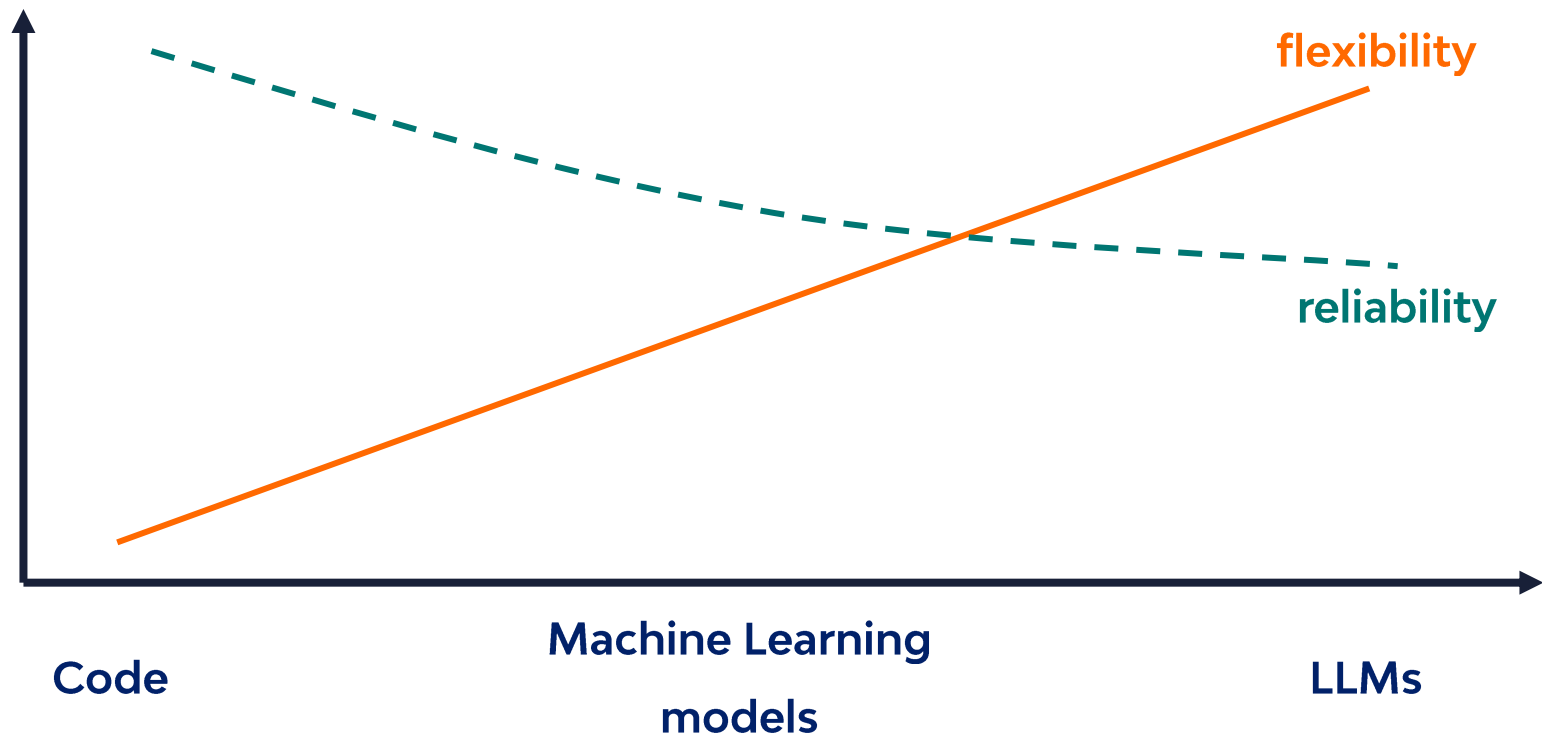
The tradeoff between flexibility and reliability



Our job is to design systems that are flexible when they need to be, and robust when they do not

Can we make GenAI systems reliable?

Flexibility vs. Reliability



There are many good practices we should use to maximize the quality and relevance of the responses we get

Prompt engineering and next steps

- Build specific and well-structured prompts

Create a presentation for a
GenAI workshop

role
task
structure



You are a GenAI expert
Create a presentation for a workshop on
the topic for a business analytics company
Use markdown to format the text

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Prompt engineering and next steps

- Build specific and well-structured prompts
- Provide examples to the model on how to do the task

You are a GenAI expert
Create a presentation for a workshop on
the topic for a business analytics company
Use markdown to format the text

prompt
examples



You are a GenAI expert
Create a presentation for a workshop on the topic for
a business analytics company
Use markdown to format the text
Here are some material we have used in the past: (...)

There are many good practices we should use to maximize the quality and relevance of the responses we get

Prompt engineering and next steps

- Build specific and well-structured prompts
- Provide examples to the model on how to do the task
- Use system, assistant and user messages

System: You are an AI assistant who is proficient on GenAI and whose job is to help the user with his tasks.

User: Create a presentation for a workshop on the topic for a business analytics company. Use markdown to format the text

Assistant (generated): Certainly, here is a (...)

User: Ok, can you move chapter 3 to the end?

Assistant (generated): (...)

There are many good practices we should use to maximize the quality and relevance of the responses we get

Prompt engineering and next steps

- Build specific and well-structured prompts
- Provide examples to the model on how to do the task
- Use system, assistant and user messages
- Divide the problem into smaller steps

Which topics should I focus on for a workshop on GenAI?

Can you provide me with a structure of the presentation, and which topics to cover in each section?

Generate the first section for me. Make sure you cover the topics bla bla bla...

(...)

I have this presentation, do I have any blind spots?



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Solving complex tasks reliably

GenAI systems rely on two types of LLMs, Chat Completion and Embedding models

The work horses of this story

Base Models

Chat Completion models will **return the most likely** sentences after a user query. This is very similar to what you have on ChatGPT

Chat Completions

What is the capital of Spain?



Madrid

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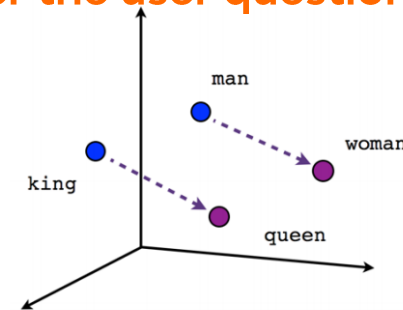
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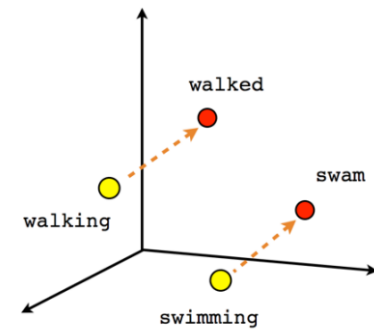
[0.3, -1.2, 2.5, -0.8, ...]

Embeddings

Embedding models capture the semantics of the user query and convert it into a vector. **They do not answer the user question**



Male-Female



Verb tense

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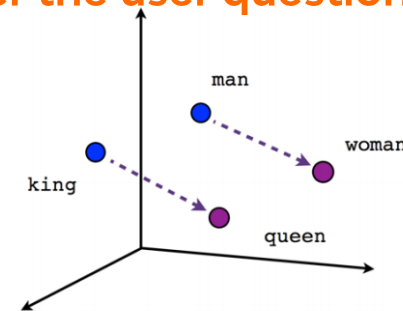
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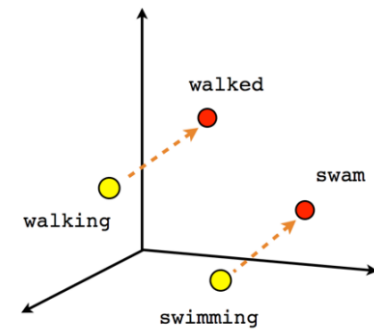
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Male-Female



Verb tense

The history of the master algorithm...

prompt engineer

noun [C] • COMPUTING • specialized

UK /'prɒpt en.dʒɪ.niə/ US /'prɑːmpt en.dʒɪ.nɪr/

[Add to word list](#)

someone who designs prompts (= instructions given to an artificial intelligence by a human using natural language rather than computer language) that will give the best possible results or answers:

With these models we can create different agents that serve as building blocks of more complex systems

The main characters of this story

Agents

Add memory



What is the capital of Spain?



Madrid

And Portugal?



Lisboa

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What is the capital of Spain?



Madrid

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Lisboa



Summarize



And France?



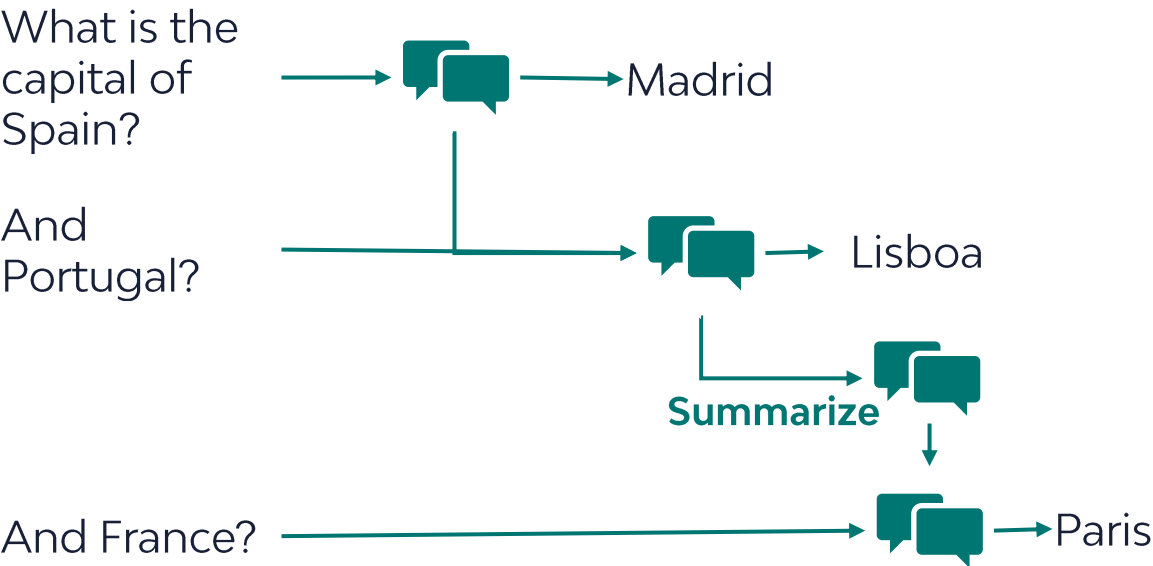
Paris

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Structured Outputs



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Tool calling agent



What is the capital of Spain?

Tools:

`Get_capital(country)`



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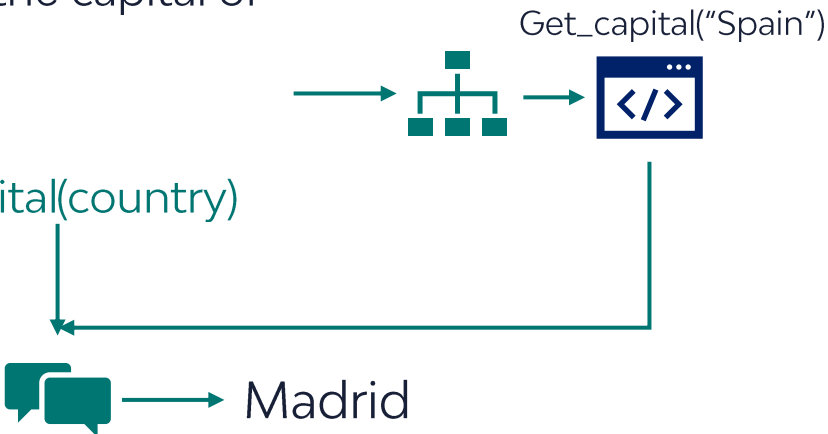
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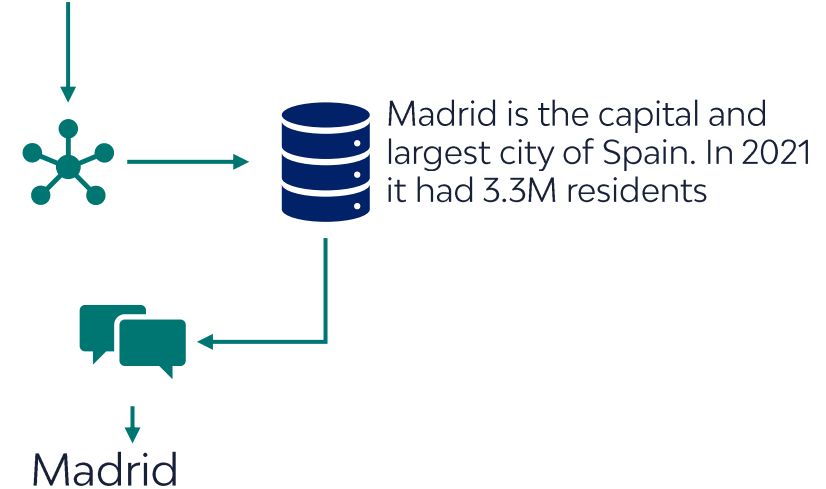
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Retrieval Aug. Generation



What is the capital of Spain?



From relatively simple models, interacting with each other, we can reach complex behaviour without sacrificing reliability

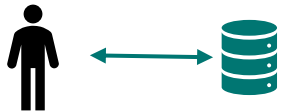
The story

Multi-agent, complex, reliable and flexible systems

Answer a user question based on data internal to the organization

When the awesome product 300 was produced for the first time?

According to the internal launches report of 2018, we started producing awesome product 300 in Jan 2018



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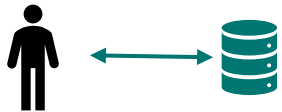
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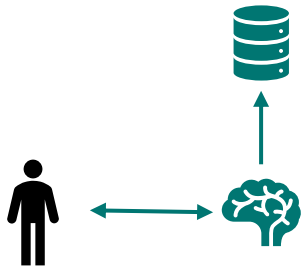


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And where?

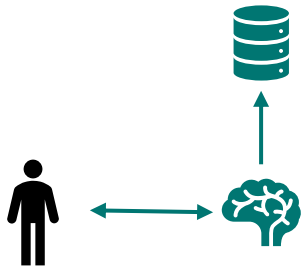
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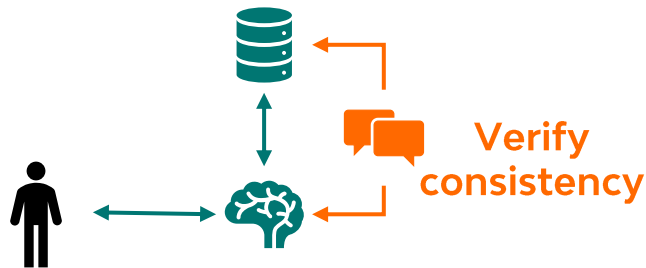
How can we avoid hallucination responses?

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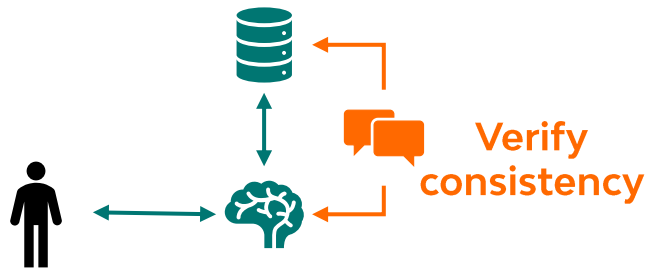
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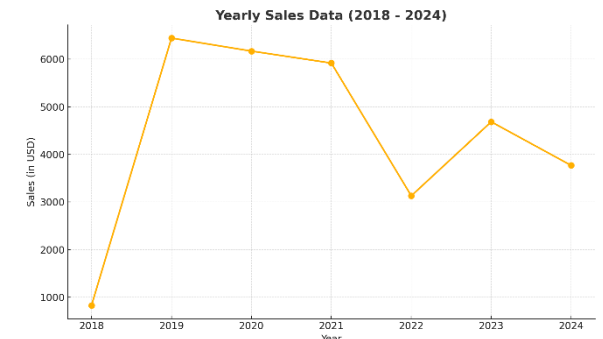
And where?

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Can you tell me how many units we have sold over the years?

Sure, here is a plot.

How can integrate structured sources of information?

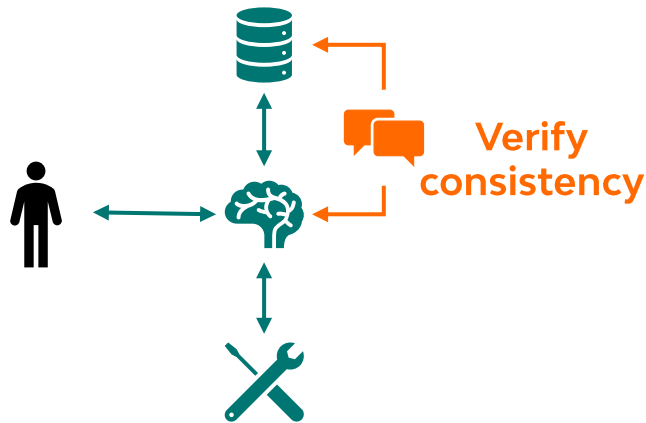


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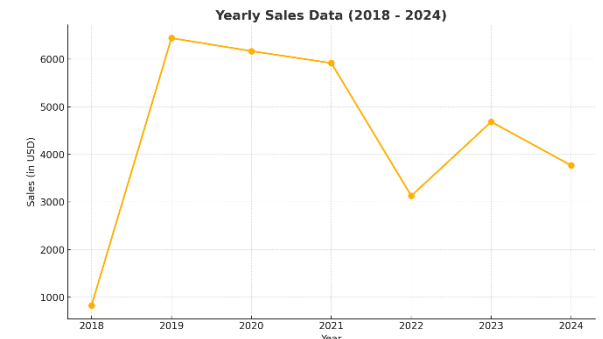
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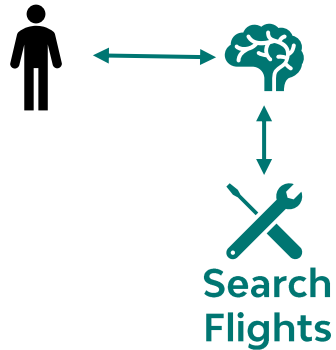


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The story

Multi-agent, complex, reliable and flexible systems

Buying tickets online



Can you buy me a ticket to London?

Certainly, what are the dates?

29 Dez to 5 Jan?

The cheapest is Rayanair123 at 235EUR.

Alright, buy those ones and send them to me.

Trying to buy ticket Rayanair123 (29/Dez 5/Jan) for 235EUR

Accept

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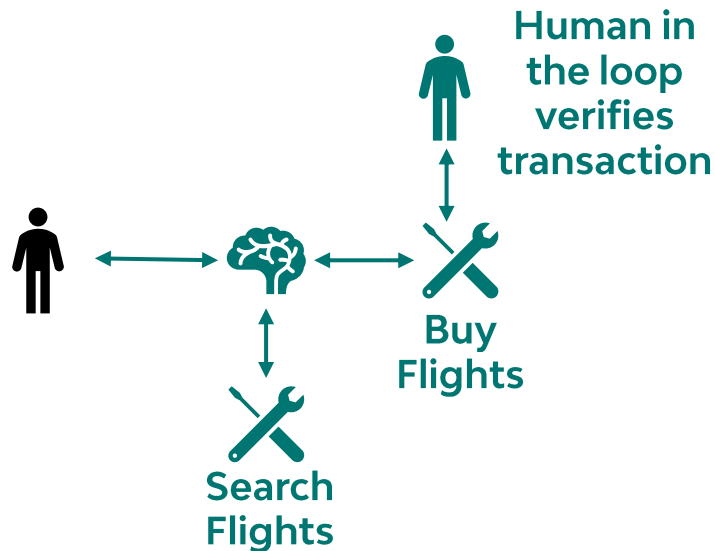
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Real use cases inside LTPlabs

A Knowledge Management tool that saves your team thousands of hours looking for information

Finding and actioning information more easily – GenBrain



Finding the info by asking questions

Connect to real-time data

GenBrain is connected to your file databases, allowing you to query data in real-time with no updates.

Find documents in seconds

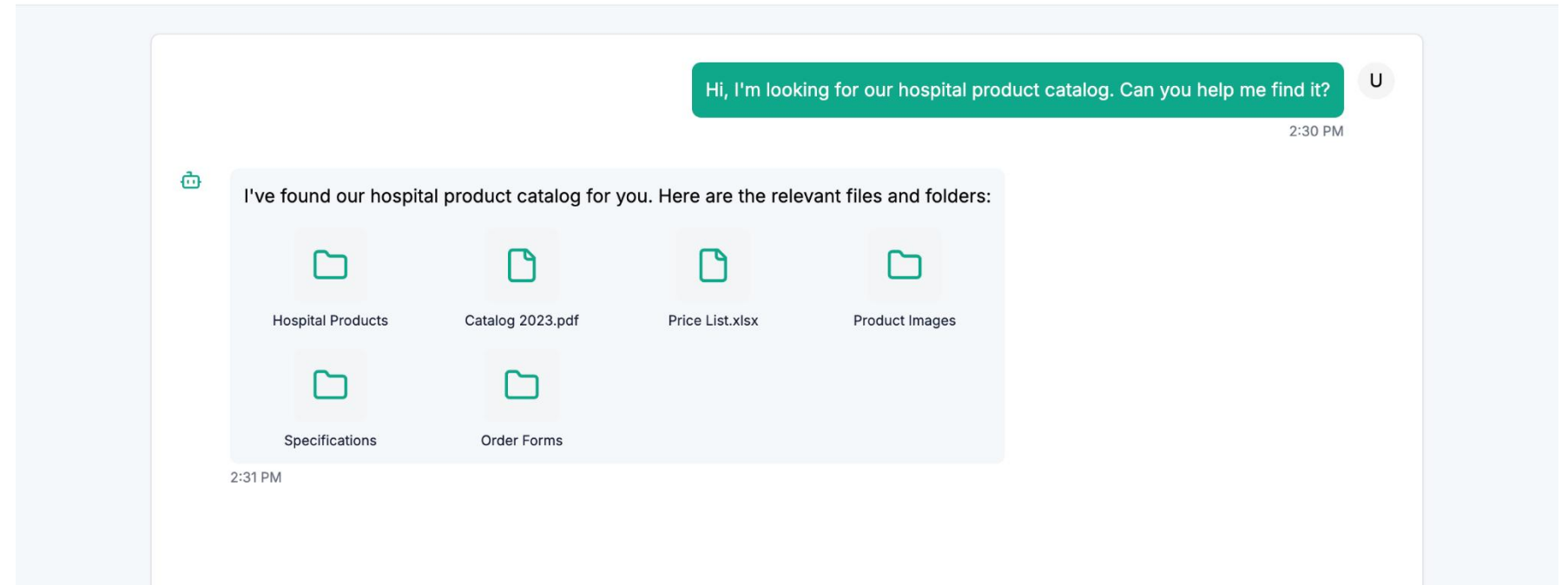
Instead of scouring through thousands of files, your team asks GenBrain for a type of document and it's instantly delivered.

Find the right information

Our data structuring techniques can maximize the accuracy for GenBrain, allowing your teams to look at fresh, up-to-date right info.

We can also input controls for information that is vetted from sources you trust.

Ready to use in one month!



An insights tool that provides instant access to data and accelerates decisions

Finding and actioning information more easily - GenDash

GenDash

Getting numbers by asking questions (what)

Increase data accessibility

Gives instant instant access to the data to anyone in your company or team.

Fast data-driven decision-making

Insights are instantly generated, skipping the analysis time required to produce them, allowing for any formatting (tables, charts, maps).

Connect to real-time data

GenDash is connected to your databases, allowing you to query data in real-time with no updates.

Adapted and customized

We can develop features based on customer's specific needs and fine-tune the model to the most valuable insights, while improving it with user feedback.

Ready to use in one month!



[Click here to try Gen Dash](#) (limited demo)

A scenario exploration tool that allows you to generate and quantify scenarios with increased speed

Playing with possibilities – GenScen



Generating What If scenarios with quick prompts

Flexibility in scenario exploration

Having a tool like GenScen allows your team include iteration in scenario exploration, minimizing the required effort to develop each one.

Lightning-fast decisions

Generating new scenarios gets so accelerated that you can make faster decisions as each of them can be properly quantified and tested before deploying.

Connect to real-time data

GenBrain is connected to your file databases, allowing you to query data in real-time with no updates.



GenScen

Baseline

Generated

Baseline Scenario

Key metrics for the scenario

Revenue:	\$1,000,000
Costs:	\$750,000
Profit:	\$250,000
ROI:	25%

Chat with GenScen

Welcome to GenScen! I'm here to help you explore different scenarios for your analytics. What would you like to analyze today?

I'd like to explore a scenario where we increase our marketing budget by 20%.

I've generated a new scenario based on a 20% increase in the marketing budget. You can see the comparison between the baseline and the new scenario in the left panel. The generated scenario shows an increase in revenue and profit, but also in costs. Would you like me to explain the implications of this change?

Type your message...

Send

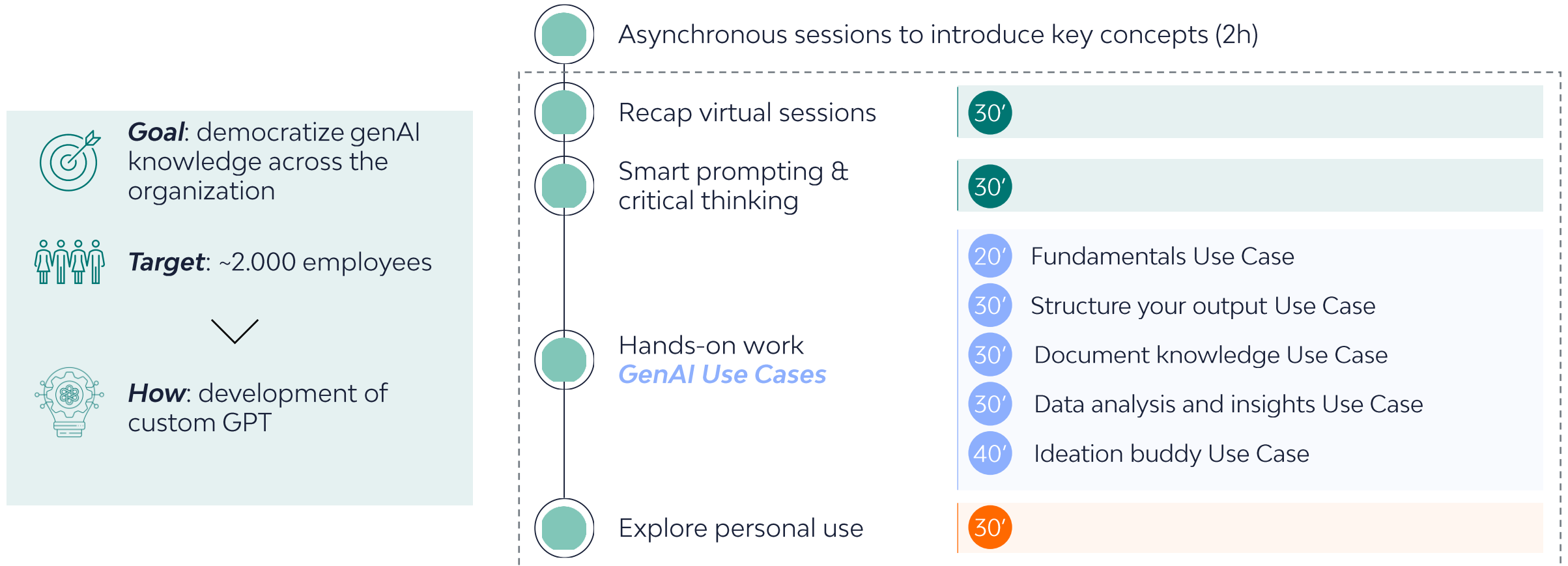
Scenario Details

Additional information about the current scenario

The generated scenario assumes a 20% increase in marketing budget.

- Projected increase in customer acquisition
- Higher brand awareness leading to more sales
- Potential for improved customer retention
- Consider the impact on different marketing channels

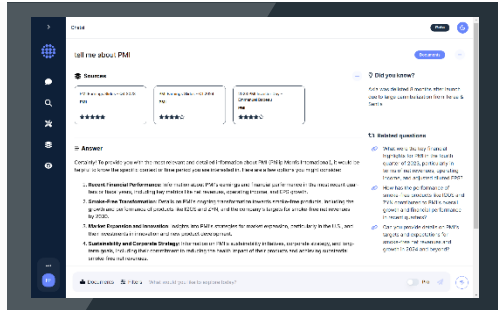
Client aims to democratize knowledge of genAI throughout the organization using a custom GPT tool



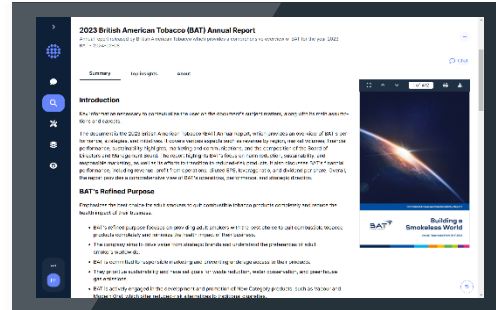
Knowledge management tool to improve productivity of +1.000 users



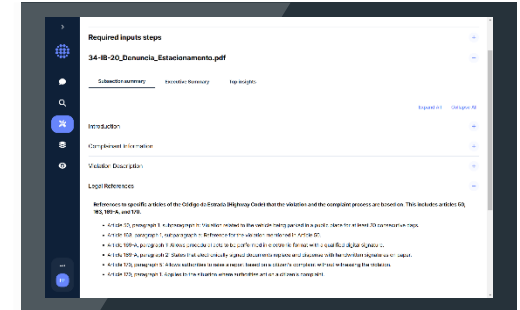
RAG pipeline to get answers on documents or data



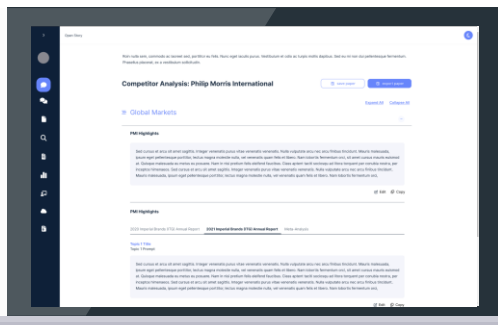
Document search to get summaries and insights



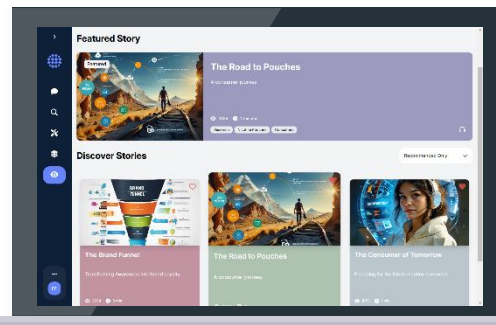
Tools for document analysis (summaries, Q&A)



Multiple document analysis for cross-doc insights



Article browser and reader for user-made stories



Key points:

- **Highly customised** use cases **tailored** for the organization
- **Custom-built Gen AI pipelines** to fit use cases and needs
- **True agile development** and support for **quick turnaround** and new **features**



Advanced Analytics &
Business Consultancy

Individuals, Teams, Organizations

GenAI will change the way we work at the individual level. Each of us now has access to a small team.

Productivity and quality opportunities with GenAI for individuals

Remove repetitive tasks

Emails, Summaries, Scheduling, Drafts, Ideation, Coding

Shift into creativity and strategy

GenAI agents at your disposal will basically work as your co-pilot through everything you do

Boost productivity

Writing, coding, preparing presentations, will be faster if not automatic

Higher standards

If making a prototype takes 2h, your client will expect a prototype.



GenAI will change the way we work at the individual level. Each of us now has access to a small team.

GenAI comes with its risk for individual work

There are **three main risks** that we need to be aware of:

- **Skill fade:** as happened with other technologies, we will lose skills. We will become worse programmers, slower at drafting papers and presentations.
Hopefully not worse thinkers
- **New skills** will be necessary, mostly regarding critical thinking and problem framing;
- **Overreliance:** GenAI will fail sometimes. Using it well and understanding the goal and structure of what we are trying to achieve is essential to minimize this risks



We will likely have GenAI agents as team members that will automate and enhance our capabilities

GenAI enables augmented collaboration in teams

Brainstorming

What is the state-of-the art for this problem? What is the typical structure when addressing X?

Content Generation

Draft meeting agendas / transcription / summaries, reports, assign tasks. Show insights on team performance and project status.

AI team members

Simple tasks where GenAI agents can perform well will likely be performed by an autonomous agent



At the organizational level, GenAI has the largest disruption potential, and the future is more uncertain

GenAI changes the way organizations operate

Operational efficiency

Better knowledge management systems, document processing, customer service, marketing, among others.

Innovation acceleration

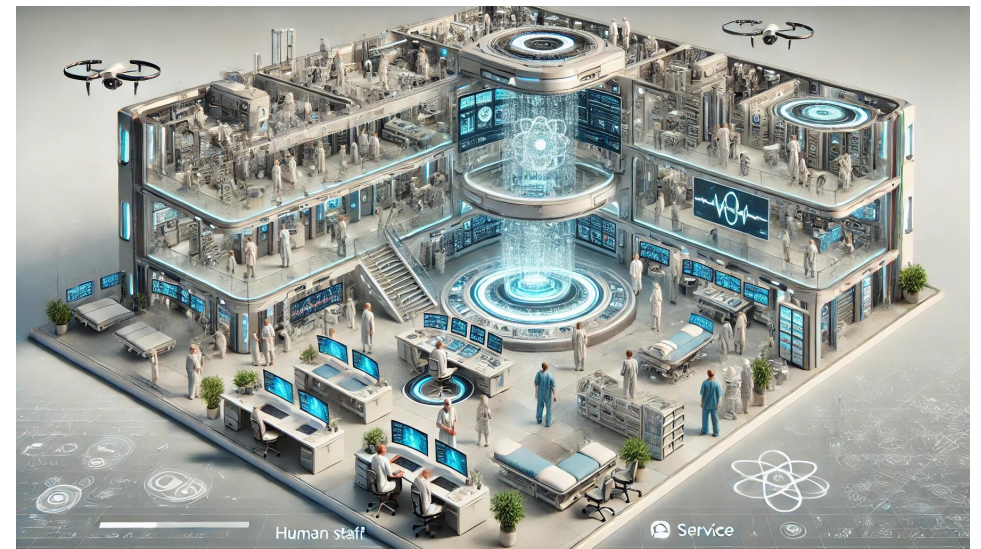
Increased productivity and more efficient teams leads to rapid iterations and drives product delivery

Culture shift

Training in GenAI tools enables individual teams to become more autonomous, proactive and productive.

AI risks

Data privacy, hallucinations, fairness and oversight will become main drivers in how organizations use AI



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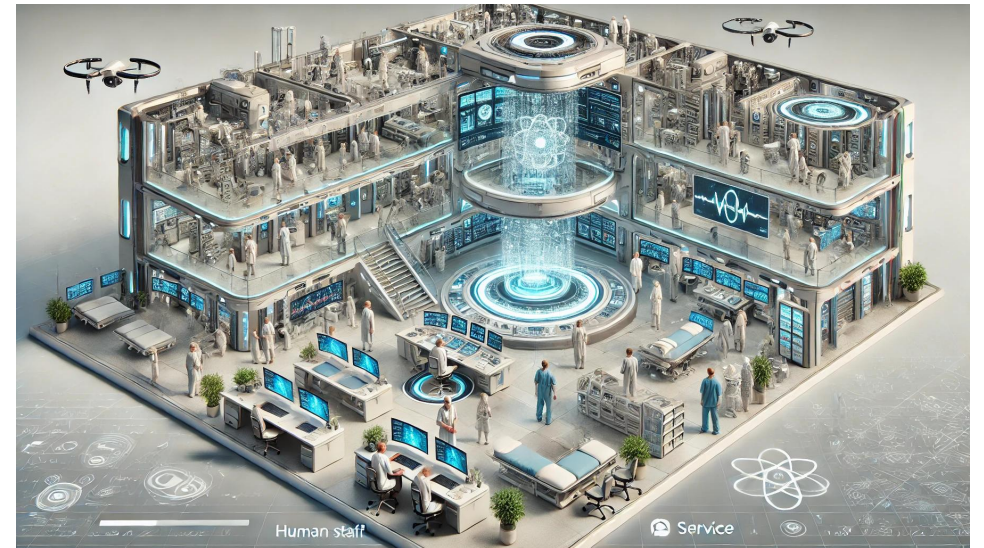
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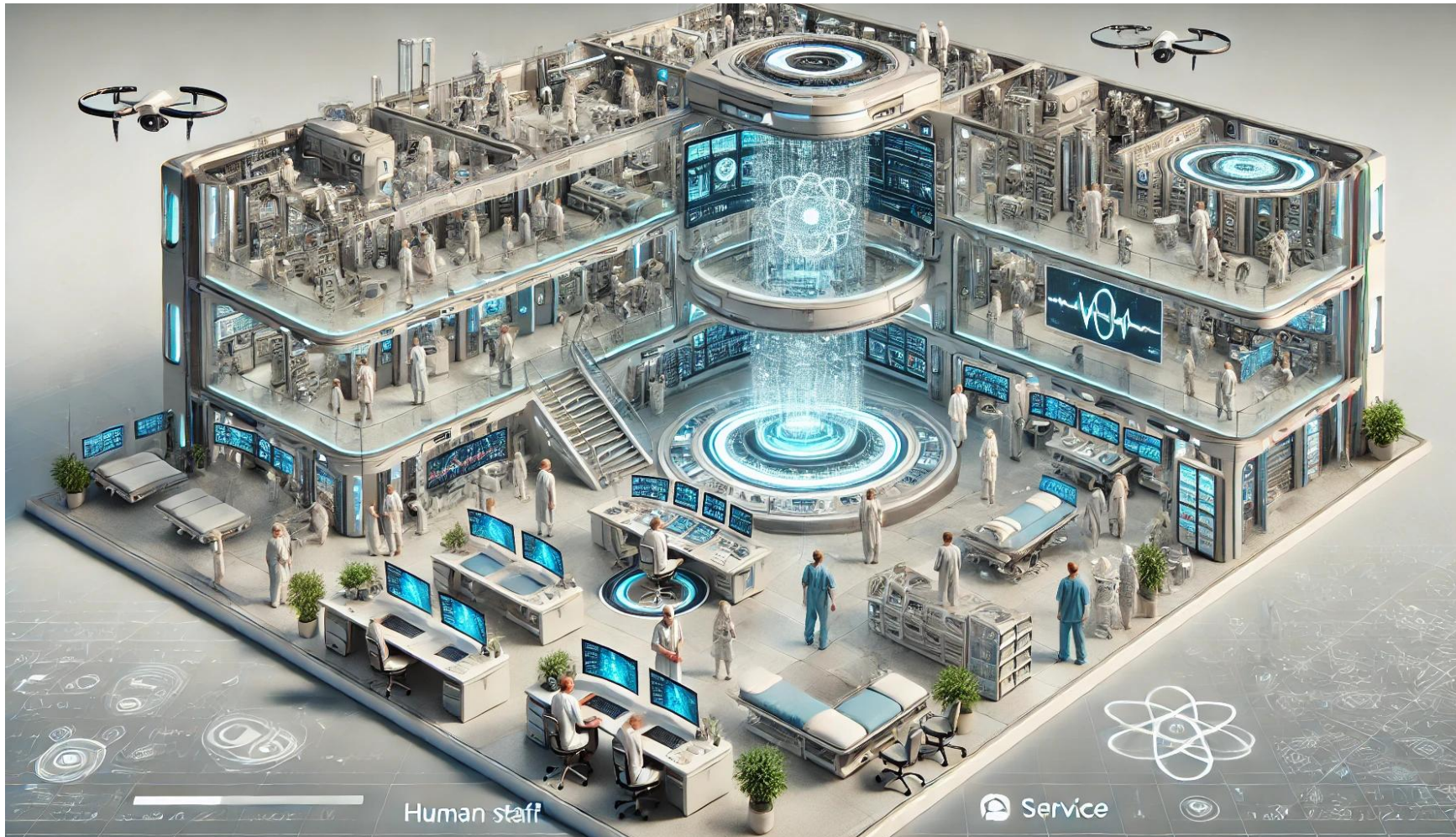
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Data privacy, hallucinations, fairness and oversight will become main drivers in how organizations use AI



One of the most remarking aspects of GenAI technology is how transversal it is across and within industries

The GenAI hospital



Retrieve patient information

Schedule consultations, exams, and surgeries

Guide patients inside the building

Fill in reports

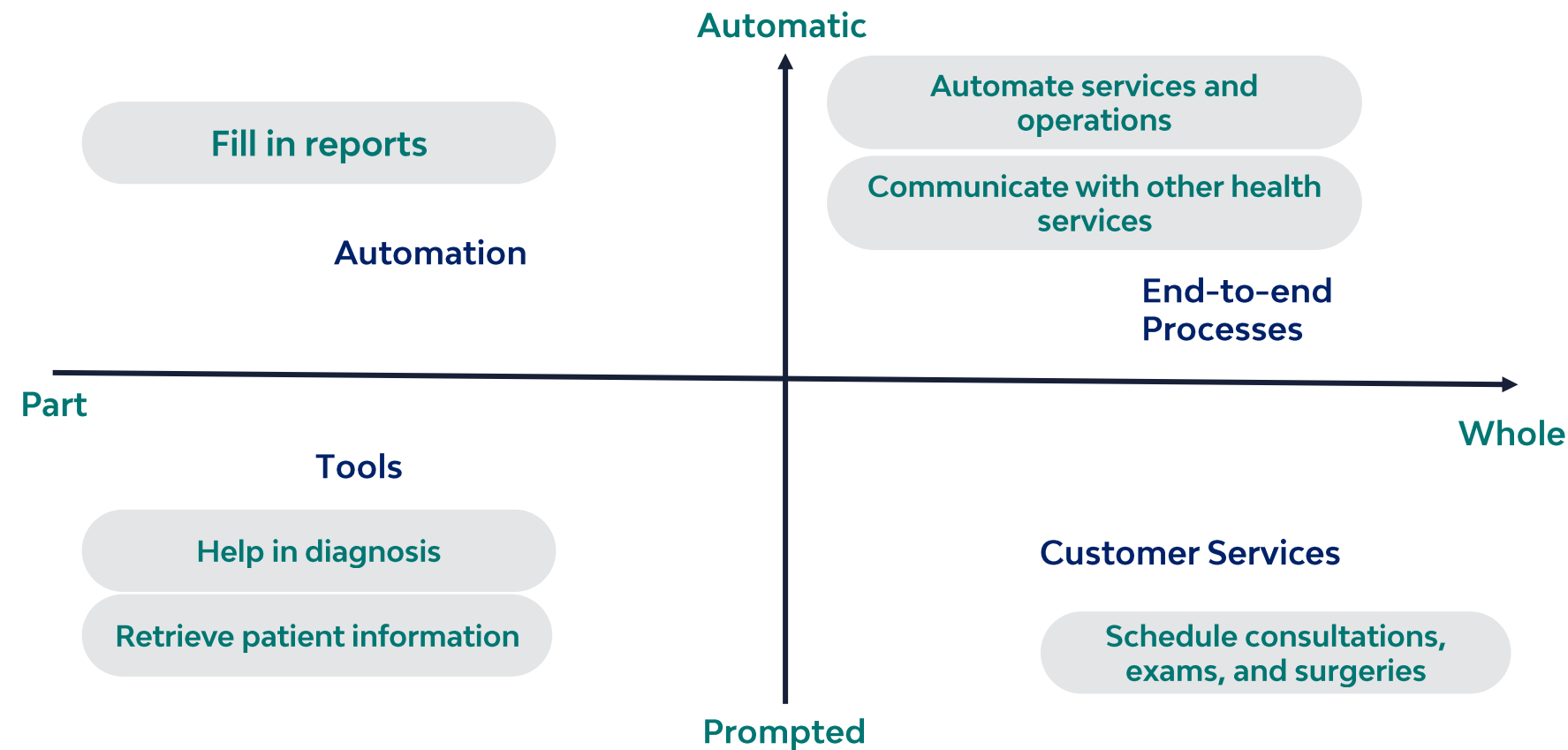
Help in diagnosis

Communicate with other health services

Automate services and operations

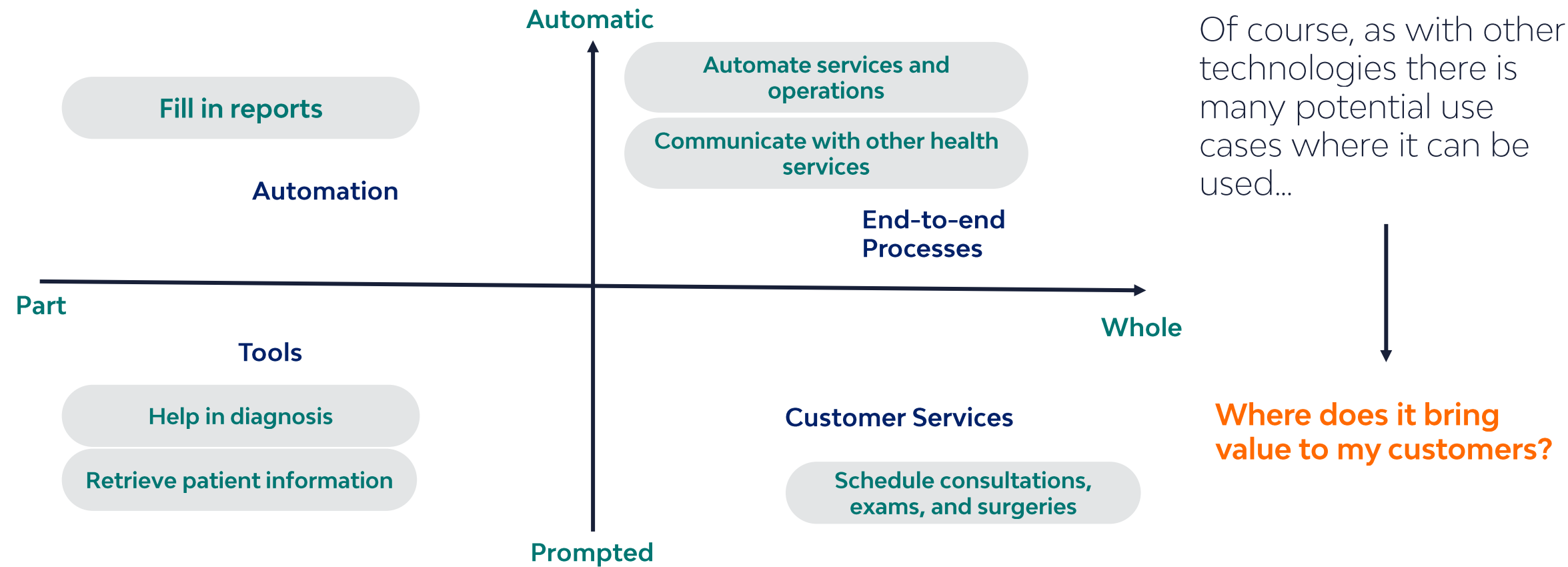
One of the most remarking aspects of GenAI technology is how transversal it is across and within industries

The interface between GenAI technology and Industry



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The interface between GenAI technology and Industry





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Thank you for your time

AI over the years

AI has come to mean different things over the years

50s – Simple logical rules: first rule-based programs

60s-70s – Expert systems: encoding in machines decision making based on if-else statements

80s – Knowledge-based systems: search, inference, reasoning, explanation

90s – Shift towards statistical methods: pattern recognition, probabilistical models

2000s – Supervised Machine Learning: learn from large datasets

2010s – Deep Learning: Deep neural networks (CNNs, RNNs) with different modalities (image, NLP, ...)

2020s – Generative AI: Large Language Models (LLMs) for multimodal data

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And 2030s?